Modules – Boards

Menu: Modules > Boards

Lists every board RateController can hear, and is where a board is given its module number and description. Each board is known by its built-in hardware (MAC) number, so boards are told apart even when they are the same type or share a module number. The list updates every second.

Board List

Column	Description
Module	The board's module number, 0–7, or **New** for a board that has no module yet.
Board	The board type, the last part of its MAC number, and its description, e.g. *ESP Rate ...0B:B0:12 Left Tank*.
State	**Connected** — the board is reporting. **Duplicate** — two or more boards are on the same module. Each is listed on its own row. Move one of them to a free module. **Update firmware** — a board on this module has firmware too old for this version of RateController. The module is not used until the firmware is updated. **New** — a new or blank board waiting for a module. New boards are listed last. **→ 3, not sent** or **Not sent** — a saved change that goes out with the next Send.

Settings

Select a board in the list to change it.

Setting	Description
Module	The module the board serves. The list holds the board's own module and the free modules only: a module that already has a board, including one that needs a firmware update, is not offered.
Description	A name for the board, up to 16 characters, e.g. "Left Tank". It is stored on the board and kept through firmware updates. The box can't be used for a Nano (it stores no description), for a board on a duplicate module, or for a new board until a module is chosen.
Web page link	For an ESP32 on WiFi, the address of its web page. Click it to open the page. An ESP32 on Ethernet shows *Ethernet, no web page*.

Setting Up a New Board

New and blank boards start without a module (ID 25). While one is reporting, the **Boards** button on the menu has an orange border.

One new board. When RateController hears one new board and no other boards, it asks "*New board found: ESP Rate ...0B:B0:12. Set it up as module 0?*" Press **Yes**. The board gets module 0 and the starting configuration for its type, then restarts and reconnects as module 0. Check the Config, Pins, Relay Pins and Valves pages against your own board afterwards.

- The question is asked once per board each time RateController runs. If you answer No, set the board up from this page instead.

- It isn't asked when other boards are connected, when two or more new boards are reporting, or while a module page has unsaved changes.

Several boards, or a module other than 0:

1. Connect the board. An ESP32 on WiFi is set up from **Modules > Communication > WiFi** instead, which gives it its module at the same time.
2. Open **Modules > Boards** and select the board marked **New**.
3. Choose a free **Module**, add a **Description** if you like, and press **Save**. The module pages switch to the chosen module.
4. RateController offers the starting configuration for the board's type (Nano, Teensy or ESP32). Press **OK** to load it, then check the Config, Pins, Relay Pins and Valves pages against your own board.
5. Press **Send** ⬆ . The board takes its module number and settings, restarts, and reconnects as that module.

Set up two new boards one at a time: Send the first and wait for it to reconnect before saving the second.

A New Board on Another Network

A new board on Ethernet starts on network 192.168.1. RateController hears it even when this PC is on a different network, and the State column shows **New, network 192.168.1**. When it's set up, with the question above or with Send, the board is also moved to RateController's module network.

- If this PC also has a connection on the board's network, RateController asks "*Move the board to network 192.168.10? It is on 192.168.1.*" Answer **No** to keep it where it is, then set RateController's module network to match on **Modules > Communication**.
- Otherwise the board is moved without asking, because RateController couldn't reach it on its old network.
- An ESP32 set up over its own hotspot isn't asked. Its hotspot is renamed with its new module (e.g. *RateModule_0_...*) and the PC joins the new name by itself. For a hotspot with a password, a message says which hotspot to join.

Moving a Board or Changing its Description

1. Select the board, choose the new **Module** and/or type a new **Description**, and press **Save**. The State column shows the change as *not sent*. A module change also moves the module pages to the new module, whose settings will be sent to the board.
2. Press **Send**. A board with a new module restarts on that module with its settings. When only descriptions have changed, Send sends just the descriptions: the boards don't restart, and the module pages don't need to be on that board's module.

- Only one module change can wait for Send at a time. Descriptions can be saved for any number of boards.
- To undo a saved module change, select the board's own module again and press Save. The module pages go back to the board's module.
- Another board can't be selected while the current one has unsaved edits. Press Save or Cancel first.

When Send is Refused

Send does not go out, and a message says why, when:

- the board being moved has stopped reporting, or another board has appeared on its new module;
- the module being edited has no board connected, or its board needs a firmware update (only when module settings are being sent; a description-only Send doesn't check this);
- a module page has unsaved changes after a module change ("*Save or cancel the pending changes first.*"). Save or cancel them and press Send again.

Except for the last case, saved module changes and descriptions are dropped when Send is refused, so they can't go out later by surprise. Save them again when the board is ready.

> **Note:** A Nano has no description. It is shown by its MAC number.

> **Note:** A board that is switched off isn't seen, so its module looks free. Make sure every board is powered before moving boards around.

> **Note:** A board with older firmware may not appear here at all. An older Nano, for example, isn't listed and its module shows as not connected. Update the board's firmware, then it appears as its module or as New. See **Updating Module Firmware** on the Help page.

Controls

Control	Action
Cancel	Discard the changes to the selected board
Save	Hold the changes until the next Send
Send	Send saved board changes, then the settings of the module being edited